MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 17: Tuesday 13 October 2026

(Week 9)

Remark. Q15.8 is a leftover exercise from Week 8.

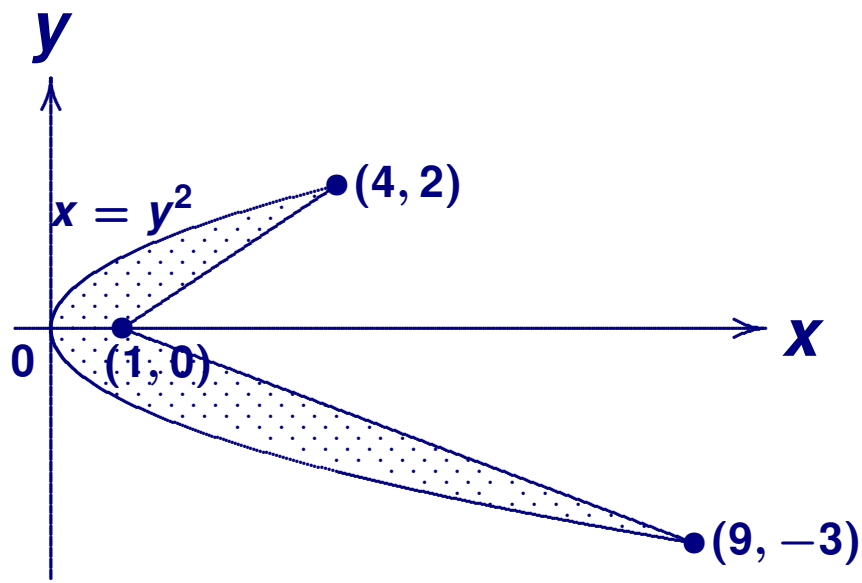
Remark. Q15.9 is supplementary exercise on double integrals (Weel 8).

Remark. Q17.1 to Q17.4 are on Section 14.8 of Thomas' Calculus.

Q15.8. Evaluate the following integral by reversing the order of integration.

$$\int_0^{\pi} \int_x^{\pi} \frac{\sin y}{y}\, dy\, dx.$$

Q15.9. (Supp. Exercise) Evaluate the double integral $\iint_D 10x\,dA$ where D is the shaded region as shown below. (The boundary of D consists of the line segment joining the two points $(1, 0)$ and $(4, 2)$, the line segment joining the points $(1, 0)$ and $(9, -3)$, and the part of the parabola $x = y^2$ joining the points $(9, -3)$ and $(4, 2)$.)



[Suggestion: Compute the double integral as a sum of two double integrals over subregions.]

Q17.1. Consider the transformation from the uv-plane to the xy-plane is given by the equations:

$$x = \frac{u - v}{\sqrt{2}}, \quad y = \frac{u + v}{\sqrt{2}}.$$

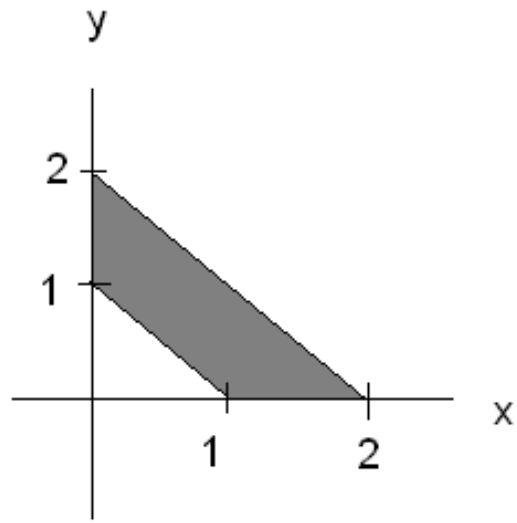
Suppose the image of a region G in the uv-plane under the above transformation is the region $R = \{(x, y) : 0 \leq x \leq 1, \ \ 0 \leq y \leq 1\}$ in the xy-plane. Find G.

[Remark: First determine the boundaries (boundary lines) of the region R. Then get the corresponding lines in the uv-plane. Then describe G by inequalities involving u and v.]

Q17.2. Evaluate

$$\iint_R \cos\left(\frac{y-x}{y+x}\right)\, dA,$$

where R is the trapezoidal region with vertices $(1,0)$, $(2,0)$, $(0,2)$ and $(0,1)$.



Suggestion: (i) Set $u = y - x$ and $v = y + x$.

(ii) First determine the corresponding region G in the uv-plane as in Q17.1.

Q17.3. Evaluate the integral $\iint_R (x^2 + 2xy)\, dA$ where R is the region bounded by the lines $y = 2x + 3$, $y = 2x + 1$, $y = 5 - x$ and $y = 2 - x$.

[Suggestion: From the equations of the boundary lines of R, get a transformation $u = u(x, y)$, $v = v(x, y)$, so that the region G in the uv-plane will be described by very simple inequalities involving u, v.]

Q17.4. Evaluate $\iint_\Omega (x^2 + y^2)\, dA$, where Ω is the region in the xy-plane bounded by the lines

$$x + y = 0, \quad x + y = 1, \quad 2x - y = 0, \quad 2x - y = 1.$$

[Suggestion: Make a change of variables.]